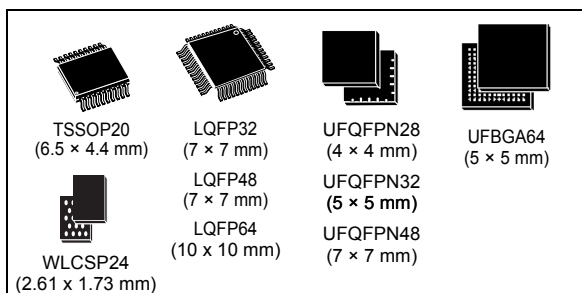


Arm[®]Cortex[®]-M0+ 32-bit MCU, 256 KB flash, 36 KB RAM,
4 x USART, timers, ADC, FDCAN, comm. I/Fs, 2-3.6V

Datasheet - production data

Features

- Includes ST state-of-the-art patented technology
- Core: Arm[®] 32-bit Cortex[®]-M0+ CPU, frequency up to 48 MHz
- 40°C to 85°C/105°C/125°C operating temperature
- Memories
 - Up to 256 Kbytes of flash memory with protection and securable area
 - Up to 36 Kbytes of SRAM with hardware parity check
- CRC calculation unit
- Reset and power management
 - Voltage range: 2.0 V to 3.6 V
 - Power-on / power-down reset (POR/PDR)
 - Programmable brownout reset (BOR)
 - Low-power modes: Sleep, Stop, Standby, Shutdown
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz crystal oscillator with calibration
 - Internal 48 MHz RC oscillator (±1 %)
 - Internal 32 kHz RC oscillator (±5 %)
- Up to 61 fast I/Os
 - All mappable on external interrupt vectors
 - All 5 V-tolerant
- 7-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 19 ext. channels)
 - Conversion range: 0 to 3.6 V
- 10 timers: 16-bit for advanced motor control, one 32-bit timer and five 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm



- Communication interfaces
 - Two I²C-bus interface supporting Fast-mode Plus (1 Mbit/s) with extra current sink; one supporting SMBus/PMBus[™] and wake-up from Stop mode
 - Four USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wake-up feature
 - Two SPIs (24 Mbit/s) with 4- to 16-bit programmable bitframe, one multiplexed with I²S interface; four extra SPIs through USARTs
- One FDCAN controller (STM32C092xx only)
- Development support: serial wire debug (SWD)
- 96-bit unique ID
- SESIP3 and PSA Level 3 target certification
- All packages ECOPACK 2 compliant

Table 1. Device summary

Reference	Part number
STM32C091xx	STM32C091FB, STM32C091GB, STM32C091KB, STM32C091CB, STM32C091RB, STM32C091FC, STM32C091EC, STM32C091GC, STM32C091KC, STM32C091CC, STM32C091RC
STM32C092xx	STM32C092FB, STM32C092GB, STM32C092KB, STM32C092CB, STM32C092RB, STM32C092FC, STM32C092EC, STM32C092GC, STM32C092KC, STM32C092CC, STM32C092RC